LINEAR ALGEBRA & CALCULUS

(23A54101)

UNIT - IV

Partial Differentiation & Applications

Topics covered:

- Functions of several variables: continuity and differentiability
- Partial derivatives, total derivative, chain rule
- Directional derivative
- Taylor's and Maclaurin's series of functions of two variables
- Jacobians and functional dependence
- Maxima and minima of functions of two variables
- Method of Lagrange multipliers

Handwritten Notes - B.Tech (JNTUA R23)

1. Functions of Several Variables – Limit, Continuity

$z = f(x, y)$ is a function of two independent variables. LIMIT: $\lim f(x, y) = L$ as $(x, y) \to (a, b)$ means f approaches L along EVERY path to (a, b).

- If the limit has different values along two paths, the limit does NOT exist.
- f is CONTINUOUS at (a, b) if $\lim f(x, y) = f(a, b)$.
- Standard paths: $y = mx$, $y = mx^2$, $x = my^2$, $x = 0$, $y = 0$.

Problem 1: discuss the continuity of $f(x, y) = xy/(x^2 + y^2)$, $f(0, 0) = 0$ at the origin.

Along $y = mx$: $f = m\,x^2/(x^2(1 + m^2)) = m/(1 + m^2)$ – depends on m ($m = 0 \to 0$; $m = 1 \to 1/2$). The limit is not unique $\Rightarrow$ lim does not exist $\Rightarrow$ f is DISCONTINUOUS at $(0, 0)$.

Problem 2: examine the continuity of $f(x, y) = x^2y/(x^2 + y^2)$, $f(0, 0) = 0$ at $(0, 0)$.

$|f| = x^2|y|/(x^2 + y^2) \le |y|$ (since $x^2 \le x^2 + y^2$) $\to 0$ as $(x, y) \to (0, 0)$. So $\lim f = 0 = f(0, 0) \Rightarrow$ CONTINUOUS at the origin.

Problem 3: show that $\lim xy^2/(x^2 + y^4)$ does not exist at $(0, 0)$.

Along $y = mx$: $m^2x^3/(x^2 + m^4x^4) = m^2x/(1 + m^4x^2) \to 0$. Along $x = y^2$: $y^4/(y^4 + y^4) = 1/2$. Different limits $\Rightarrow$ the limit does not exist (checking only straight lines is not enough).

Differentiability: f is differentiable at (a, b) if $\Delta f = f_x\Delta x + f_y\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y$ with $\varepsilon_1, \varepsilon_2 \to 0$. If f_x and f_y exist and are CONTINUOUS at (a, b), f is differentiable there; differentiable $\Rightarrow$ continuous (converse false).

Problem 4: $f(x, y) = \sqrt{|xy|}$ at $(0, 0)$ – partial derivatives exist but f is not differentiable.

$f_x(0,0) = \lim [f(h,0) - f(0,0)]/h = 0$ and $f_y(0,0) = 0$. If f were differentiable, $\Delta f = 0\cdot h + 0\cdot K +$ small terms, i.e. $\sqrt{|hK|} = o(\sqrt{(h^2+K^2)})$; but along $h = K$: $\sqrt{(h^2)} = |h| = (1/\sqrt{2})\sqrt{(2h^2)}$ – a constant multiple of the distance, not smaller. Hence NOT differentiable.

2. Partial Derivatives

$f_x = \partial f/\partial x$ is the derivative with respect to x treating y as constant (and vice-versa). Second order: $f_{xx}, f_{yy}, f_{xy} = \partial^2 f/\partial y \partial x, f_{yx}$. If f_{xy}, f_{yx} are continuous then $f_{xy} = f_{yx}$ (Schwarz / Clairaut).

Problem 5: for $u = x^4 + y^4 - 4x^2y^2$ find $u_x, u_y, u_{xx}, u_{yy}, u_{xy}$.

$$u_x = 4x^3 - 8xy^2 \,,\; u_y = 4y^3 - 8x^2y$$

$$u_{xx} = 12x^2 - 8y^2 \,,\; u_{yy} = 12y^2 - 8x^2 \,,\; u_{xy} = u_{yx} = -16xy$$

Problem 6: if $u = \log(x^2 + y^2)$ show that $u_{xx} + u_{yy} = 0$ (Laplace equation).

$$u_x = \frac{2x}{x^2+y^2} \,,\; u_{xx} = \frac{2(y^2 - x^2)}{(x^2+y^2)^2} \,,\; u_{yy} = \frac{2(x^2 - y^2)}{(x^2+y^2)^2}$$

Adding: $u_{xx} + u_{yy} = 0$ ✓.

Problem 7: if $u = \tan^{-1}(y/x)$ show that $u_{xx} + u_{yy} = 0$.

$$u_x = \frac{-y}{x^2+y^2} \,,\; u_y = \frac{x}{x^2+y^2}$$

$$u_{xx} = \frac{2xy}{(x^2+y^2)^2} \,,\; u_{yy} = \frac{-2xy}{(x^2+y^2)^2} \Rightarrow u_{xx} + u_{yy} = 0 \text{ ✓}$$

Problem 8: if $u = x^y$ show that $u_{xy} = u_{yx}$.

$u_x = y\,x^{y-1} \Rightarrow u_{xy} = x^{y-1} + y\,x^{y-1}\log x = x^{y-1}(1 + y\log x)$. $u_y = x^y \log x \Rightarrow u_{yx} = y\,x^{y-1}\log x + x^y\cdot(1/x) = x^{y-1}(1 + y\log x)$. Equal ✓.

Problem 9: if $u = \log(x^3 + y^3 + z^3 - 3xyz)$ show that $(\partial/\partial x + \partial/\partial y + \partial/\partial z)\,u = 3/(x+y+z)$ and $(\partial/\partial x + \partial/\partial y + \partial/\partial z)^2\,u = -9/(x+y+z)^2$.

Let $S = x^3 + y^3 + z^3 - 3xyz$. Sum of the three partials of $S = 3(x^2 - yz) + 3(y^2 - zx) + 3(z^2 - xy) = 3(x^2 + y^2 + z^2 - xy - yz - zx) = 3S/(x+y+z)$ [since

$S = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)]$.

$$(u_x+u_y+u_z) = \frac{3S/(x+y+z)}{S} = \frac{3}{x+y+z}$$

Applying the operator once more (each derivative of $1/(x+y+z)$ is $-1/(x+y+z)^2$):

$$\boxed{(\partial_x+\partial_y+\partial_z)\frac{3}{x+y+z} = -\frac{9}{(x+y+z)^2}} \checkmark$$

Euler's theorem on homogeneous functions: if $f(x,y)$ is homogeneous of degree n, then $x f_x + y f_y = n f$. Further $x^2 f_{xx} + 2xy f_{xy} + y^2 f_{yy} = n(n-1) f$.

Problem 10: if $u = \sin^{-1}[(x^2+y^2)/(x+y)]$ show that $x u_x + y u_y = \tan u$.

$\sin u = (x^2+y^2)/(x+y) =: f$, homogeneous of degree 1. By Euler: $x f_x + y f_y = f$. But $f = \sin u \Rightarrow f_x = \cos u \cdot u_x$, $f_y = \cos u \cdot u_y$:

$$\boxed{\cos u\,(x u_x + y u_y) = \sin u \Rightarrow x u_x + y u_y = \tan u}$$

Problem 11: verify Euler's theorem for $u = x^3+y^3+3x^2y$ (degree 3).

$u_x = 3x^2+6xy$, $u_y = 3y^2+3x^2$. $x u_x + y u_y = 3x^3+6x^2y+3y^3+3x^2y = 3(x^3+y^3+3x^2y) = 3u$ ✓.

3. Total Derivative and Chain Rule

Total differential: $dz = (\partial z/\partial x)\,dx + (\partial z/\partial y)\,dy$.

Total derivative: if $z = f(x,y)$ with $x = x(t)$, $y = y(t)$:

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}}$$

Chain rule (two parameters): $z = f(x,y)$, $x = x(u,v)$, $y = y(u,v)$:

$\partial z/\partial u = z_x x_u + z_y y_u$, $\partial z/\partial v = z_x x_v + z_y y_v$.

Implicit function: if $f(x,y) = c$ defines $y(x)$: $dy/dx = -f_x/f_y$.

Problem 12: if $z = x^2y + xy^2$ with $x = at^2$, $y = 2at$, find dz/dt.

$z_x = 2xy + y^2$, $z_y = x^2 + 2xy$; $dx/dt = 2at$, $dy/dt = 2a$.

$$\frac{dz}{dt} = (2xy + y^2)(2at) + (x^2 + 2xy)(2a)$$

Substituting $x = at^2$, $y = 2at$: $2xy + y^2 = 4a^2t^3 + 4a^2t^2$; $x^2 + 2xy = a^2t^4 + 4a^2t^3$.

$$\frac{dz}{dt} = 8a^3t^4 + 8a^3t^3 + 2a^3t^4 + 8a^3t^3 = 10a^3t^4 + 16a^3t^3$$

Problem 13: if $u = x^2 + y^2 + z^2$ and $x = e^t$, $y = e^t \sin t$, $z = e^t \cos t$, find du/dt.

Directly: $u = e^{2t}(1 + \sin^2 t + \cos^2 t) = 2e^{2t} \Rightarrow du/dt = 4e^{2t}$. By the chain rule: $2x \cdot e^t + 2y \cdot e^t(\sin t + \cos t) + 2z \cdot e^t(\cos t - \sin t) = 4e^{2t}$ ✓.

Problem 14: if $z = f(x, y)$, $x = u \cos \alpha - v \sin \alpha$, $y = u \sin \alpha + v \cos \alpha$ show that $z_x^2 + z_y^2 = z_u^2 + z_v^2$.

$x_u = \cos \alpha$, $x_v = -\sin \alpha$, $y_u = \sin \alpha$, $y_v = \cos \alpha$. $z_u = z_x \cos \alpha + z_y \sin \alpha$, $z_v = -z_x \sin \alpha + z_y \cos \alpha$. Squaring and adding:

$$z_u^2 + z_v^2 = z_x^2(\cos^2\alpha + \sin^2\alpha) + z_y^2(\sin^2\alpha + \cos^2\alpha) = z_x^2 + z_y^2 \checkmark$$

Problem 15: if $z = f(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$, show that $z_x^2 + z_y^2 = z_r^2 + (1/r^2) z_\theta^2$.

$z_r = z_x \cos \theta + z_y \sin \theta$; $z_\theta = -r z_x \sin \theta + r z_y \cos \theta$. Then $z_r^2 + z_\theta^2/r^2 = (z_x \cos \theta + z_y \sin \theta)^2 + (-z_x \sin \theta + z_y \cos \theta)^2 = z_x^2 + z_y^2$ ✓.

Problem 16: find dy/dx if $x^3 + y^3 = 3axy$.

$f_x = 3x^2 - 3ay$, $f_y = 3y^2 - 3ax$.

$$\frac{dy}{dx} = -\frac{f_x}{f_y} = \frac{ay - x^2}{y^2 - ax}$$

4. Directional Derivative

The derivative of f in the direction of a unit vector $\hat{u} = (l, m, n)$:

$$D_{\hat{u}} f = \nabla f \cdot \hat{u} = l f_x + m f_y + n f_z \text{ where } \nabla f = (f_x, f_y, f_z)$$

- Greatest rate of increase of f at a point is $|\nabla f|$, in the direction of ∇f.
- Smallest (most negative) rate is $-|\nabla f|$ in the direction $-\nabla f$. Along a level curve $D_{\hat{u}} f = 0$.
- The vector $\hat{u}$ must be a UNIT vector – divide a given direction by its length first.

Problem 17: find the directional derivative of $f = x^2 + xy + y^2$ at $(1, 2)$ in the direction of the vector $i + j$.

$\nabla f = (2x + y, x + 2y) = (4, 5)$ at $(1, 2)$. $\hat{u} = (1, 1)/\sqrt{2}$.

$$D_{\hat{u}} f = (4 + 5)/\sqrt{2} = 9/\sqrt{2} \approx 6.364$$

Maximum rate of change $= |\nabla f| = \sqrt{(16 + 25)} = \sqrt{41}$ in the direction $(4, 5)$.

Problem 18: find the directional derivative of $f = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2i - j - 2K$.

$\nabla f = (2xyz + 4z^2, x^2z, x^2y + 8xz) = (8, -1, -10)$ at $(1, -2, -1)$. $|2i - j - 2K| = 3 \Rightarrow \hat{u} = (2, -1, -2)/3$.

$$D_{\hat{u}} f = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

Problem 19: in what direction from $(1, 1)$ is the directional derivative of $f = x^2 - y^2 + 2xy$ a maximum? find its value.

$\nabla f = (2x + 2y, -2y + 2x) = (4, 0)$ at $(1, 1)$. Direction i (along the positive x-axis), maximum value $|\nabla f| = 4$.

5. Taylor's and Maclaurin's Series – Two Variables

$$f(x,y) = f(a,b) + [(x-a)f_x + (y-b)f_y]$$

$$+ \frac{1}{2!}[(x-a)^2 f_{xx} + 2(x-a)(y-b)f_{xy} + (y-b)^2 f_{yy}] + \dots$$

(all derivatives evaluated at (a, b)). With $h = x - a$, $K = y - b$ the n-th term is $\frac{1}{n!}(h\,\partial/\partial x + K\,\partial/\partial y)^n f$. Maclaurin's series: $(a, b) = (0, 0)$.

Problem 20: expand $e^x \sin y$ in powers of x and y up to the third degree.

Derivatives at $(0, 0)$: $f = 0$; $f_x = 0$; $f_y = 1$; $f_{xx} = 0$; $f_{xy} = 1$; $f_{yy} = 0$; $f_{xxy} = 1$; $f_{yyy} = -1$; others 0.

$$e^x \sin y = y + xy + \frac{x^2 y}{2} - \frac{y^3}{6} + \dots$$

(Quick check: multiply the one-variable series $(1 + x + x^2/2 + \dots)(y - y^3/6 + \dots)$.)

Problem 21: expand $e^x \log(1 + y)$ up to the third degree.

$$e^x \log(1+y) = \left(1 + x + \frac{x^2}{2}\right)\left(y - \frac{y^2}{2} + \frac{y^3}{3}\right) = y + xy - \frac{y^2}{2} + \frac{x^2 y}{2} - \frac{xy^2}{2} + \frac{y^3}{3} + \dots$$

Problem 22: expand $f(x, y) = x^2 y + 3y - 2$ in powers of $(x - 1)$ and $(y + 2)$ up to the third degree.

$f(1, -2) = -2 - 6 - 2 = -10$; $f_x = 2xy \to -4$; $f_y = x^2 + 3 \to 4$; $f_{xx} = 2y \to -4$; $f_{xy} = 2x \to 2$; $f_{yy} = 0$; $f_{xxy} = 2$; others 0.

$$f = -10 + [-4h + 4K] + \frac{1}{2}[-4h^2 + 4hK] + \frac{1}{6}[3 \cdot 2\, h^2 K]$$

$$= -10 - 4h + 4K - 2h^2 + 2hK + h^2 K \quad (h = x - 1,\ K = y + 2)$$

Problem 23: expand $\tan^{-1}(y/x)$ about $(1, 1)$ up to the second degree.

$f(1,1) = \pi/4$; $f_x = -y/(x^2+y^2) = -1/2$; $f_y = x/(x^2+y^2) = 1/2$; $f_{xx} = 2xy/(x^2+y^2)^2 = 1/2$; $f_{xy} = (y^2 - x^2)/(x^2+y^2)^2 = 0$; $f_{yy} = -1/2$.

$$\tan^{-1}(y/x) = \frac{\pi}{4} - \frac{x-1}{2} + \frac{y-1}{2} + \frac{(x-1)^2}{4} - \frac{(y-1)^2}{4} + \dots$$

Problem 24: Maclaurin series of $\cos x \cos y$ up to the fourth degree.

$$\cos x \cos y = 1 - \frac{x^2+y^2}{2} + \frac{x^2y^2}{4} + \frac{x^4+y^4}{24} + \dots$$

6. Jacobians and Functional Dependence

For $u = u(x,y)$, $v = v(x,y)$ the JACOBIAN is the determinant

$$J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{bmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{bmatrix}$$

- $J \cdot J' = 1$ where $J' = \partial(x,y)/\partial(u,v)$ (inverse transformation).
- Chain rule: $\partial(u,v)/\partial(r,s) = \partial(u,v)/\partial(x,y) \cdot \partial(x,y)/\partial(r,s)$.
- Functional dependence: u, v (or u, v, w) are functionally DEPENDENT $\Leftrightarrow J = 0$; independent $\Leftrightarrow J \neq 0$.

Problem 25: find $\partial(x,y)/\partial(r,\theta)$ for $x = r\cos\theta$, $y = r\sin\theta$; and $\partial(r,\theta)/\partial(x,y)$.

$$J = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} = r\cos^2\theta + r\sin^2\theta = r \text{ , so } \partial(r,\theta)/\partial(x,y) = 1/r$$

Problem 26: $x = u(1-v)$, $y = uv$: find $\partial(x,y)/\partial(u,v)$.

$$J = \begin{bmatrix} 1-v & -u \\ v & u \end{bmatrix} = u(1-v) + uv = u$$

Problem 27: cylindrical and spherical coordinates: find the Jacobians.

Cylindrical ($x = r\cos\theta, y = r\sin\theta, z = z$): $\partial(x, y, z)/\partial(r, \theta, z) = r$

Spherical: $x = r\sin\theta\cos\varphi$, $y = r\sin\theta\sin\varphi$, $z = r\cos\theta$:

$$\boxed{\partial(x, y, z)/\partial(r, \theta, \varphi) = r^2\sin\theta}$$

Problem 28: if $u = x^2 - y^2$, $v = 2xy$ and $x = r\cos\theta$, $y = r\sin\theta$, find $\partial(u, v)/\partial(r, \theta)$.

$\partial(u, v)/\partial(x, y) = \det\begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix} = 4x^2 + 4y^2 = 4r^2$. $\partial(x, y)/\partial(r, \theta) = r$.

$$\boxed{\partial(u, v)/\partial(r, \theta) = 4r^2 \cdot r = 4r^3}$$

Problem 29: show that $u = x + y + z$, $v = xy + yz + zx$, $w = x^3 + y^3 + z^3 - 3xyz$ are functionally dependent and find the relation.

Rows of the Jacobian: $(1, 1, 1)$; $(y + z, z + x, x + y)$; $(3(x^2 - yz), 3(y^2 - zx), 3(z^2 - xy))$. Since $w = u^3 - 3uv$ (shown below), Row 3 $= (3u^2 - 3v)\cdot$Row 1 $- 3u\cdot$Row 2, so the rows are linearly dependent and the determinant $J = 0 \Rightarrow$ DEPENDENT.

Relation: $w = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$ and $x^2 + y^2 + z^2 = u^2 - 2v$, so

$$\boxed{w = u(u^2 - 3v) = u^3 - 3uv}$$

Problem 30: prove $u = (x + y)/(1 - xy)$ and $v = \tan^{-1}x + \tan^{-1}y$ are dependent.

$u_x = (1 + y^2)/(1 - xy)^2$, $u_y = (1 + x^2)/(1 - xy)^2$; $v_x = 1/(1 + x^2)$, $v_y = 1/(1 + y^2)$. $J = u_xv_y - u_yv_x = [(1 + y^2)/(1 + y^2) - (1 + x^2)/(1 + x^2)]/(1 - xy)^2 = 0$. Relation: $\tan v = (x + y)/(1 - xy) = u$.

Problem 31: are $u = xy/z$, $v = yz/x$, $w = zx/y$ independent?

$J = 4$ (a non-zero constant) $\Rightarrow u, v, w$ are functionally independent. (Indeed $uv = y^2$, $vw = z^2$, $wu = x^2$ can be solved for x, y, z.)

7. Maxima and Minima of $f(x, y)$

Working rule:

1. Solve $f_x = 0$, $f_y = 0$ – the solutions are the STATIONARY (critical) points.
2. At each such point compute $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$.
3. $rt - s^2 > 0$ and $r < 0 \rightarrow$ MAXIMUM; $rt - s^2 > 0$ and $r > 0 \rightarrow$ MINIMUM.
4. $rt - s^2 < 0 \rightarrow$ SADDLE point (neither max nor min); $rt - s^2 = 0 \rightarrow$ doubtful, further study.

Problem 32: find the extreme values of $f = x^3 + y^3 - 3axy$ $(a > 0)$.

$f_x = 3x^2 - 3ay = 0$, $f_y = 3y^2 - 3ax = 0 \Rightarrow y = x^2/a$ and $x^4 = a^3x \Rightarrow x = 0$ or $x = a$. Points $(0, 0)$ and (a, a).

$r = 6x$, $s = -3a$, $t = 6y$. At $(0, 0)$: $rt - s^2 = -9a^2 < 0 \rightarrow$ SADDLE. At (a, a): $rt - s^2 = 36a^2 - 9a^2 = 27a^2 > 0$ and $r = 6a > 0 \rightarrow$ MINIMUM.

$$f_{min} = f(a, a) = a^3 + a^3 - 3a^3 = -a^3$$

Problem 33: find the minimum of $f = x^2 + y^2 + 6x + 12$.

$f_x = 2x + 6 = 0 \rightarrow x = -3$; $f_y = 2y = 0 \rightarrow y = 0$. $r = 2$, $s = 0$, $t = 2$: $rt - s^2 = 4 > 0$ and $r > 0 \rightarrow$ MINIMUM at $(-3, 0)$, value $f = 9 - 18 + 12 = 3$.

Problem 34: find the maximum of $f = x^3y^2(1 - x - y)$.

$f = x^3y^2 - x^4y^2 - x^3y^3$; $f_x = x^2y^2(3 - 4x - 3y) = 0$, $f_y = x^3y(2 - 2x - 3y) = 0$.

For $x, y \neq 0$: $4x + 3y = 3$ and $2x + 3y = 2 \Rightarrow x = 1/2$, $y = 1/3$.

Exact values: $r = -1/9$, $s = -1/12$, $t = -1/8$; $rt - s^2 = 1/144 > 0$ and $r < 0 \Rightarrow$ MAXIMUM.

$$f_{max} = (1/2)^3 (1/3)^2 (1 - 1/2 - 1/3) = \frac{1}{8} \cdot \frac{1}{9} \cdot \frac{1}{6} = \frac{1}{432}$$

Problem 35: show that $f = \sin x + \sin y + \sin(x + y)$ has a maximum at $(\pi/3, \pi/3)$.

$f_x = \cos x + \cos(x+y) = 0$, $f_y = \cos y + \cos(x+y) = 0 \Rightarrow \cos x = \cos y \Rightarrow x = y$; then $\cos x + \cos 2x = 0 \Rightarrow 2\cos^2 x + \cos x - 1 = 0 \Rightarrow \cos x = 1/2 \Rightarrow x = y = \pi/3$.

$r = -\sin x - \sin(x+y) = -\sqrt{3}/2 - \sqrt{3}/2 = -\sqrt{3}$; $s = -\sin(x+y) = -\sqrt{3}/2$; $t = -\sqrt{3}$.

$rt - s^2 = 3 - 3/4 = 9/4 > 0$ and $r < 0 \Rightarrow$ MAXIMUM.

$$f_{max} = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

Problem 36: divide 24 into three parts so that the product of the parts is a maximum.

Let the parts be $x, y, 24 - x - y$; $f = xy(24 - x - y)$. $f_x = y(24 - 2x - y) = 0$, $f_y = x(24 - x - 2y) = 0 \Rightarrow 2x + y = 24$, $x + 2y = 24 \Rightarrow x = y = 8$. $r = -2y = -16$, $s = 24 - 2x - 2y = -8$, $t = -2x = -16$: $rt - s^2 = 256 - 64 > 0$, $r < 0 \Rightarrow$ maximum.

parts 8, 8, 8; maximum product = 512

Problem 37: find the maximum of $f = xy(a - x - y)$.

$f_x = y(a - 2x - y) = 0$, $f_y = x(a - x - 2y) = 0 \Rightarrow x = y = a/3$ (also $(0,0)$, $(a,0)$, $(0,a)$ – saddles). $r = -2a/3$, $s = -a/3$, $t = -2a/3 \Rightarrow rt - s^2 = a^2/3 > 0$, $r < 0 \Rightarrow$ MAX. $f_{max} = (a/3)(a/3)(a/3) = a^3/27$.

8. Lagrange's Method of Multipliers

To find the extreme values of $f(x, y, z)$ subject to a constraint $\varphi(x, y, z) = 0$:

1. Form $F = f + \lambda\varphi$ (λ = Lagrange multiplier).
2. Solve $F_x = 0$, $F_y = 0$, $F_z = 0$ together with $\varphi = 0$ for x, y, z, λ.
3. Substitute the point(s) in f to get the extreme value; nature from the geometry of the problem.
4. Two constraints $\varphi_1 = 0$, $\varphi_2 = 0$: $F = f + \lambda_1\varphi_1 + \lambda_2\varphi_2$.

Problem 38: find the minimum value of $x^2 + y^2 + z^2$ subject to $x + y + z = 3a$.

$F = x^2 + y^2 + z^2 + \lambda(x + y + z - 3a)$. $F_x = 2x + \lambda = 0$, $F_y = 2y + \lambda = 0$, $F_z = 2z + \lambda = 0 \Rightarrow x = y = z = -\lambda/2$. With the constraint: $3x = 3a \Rightarrow x = y = z = a$.

$$\text{minimum value} = a^2 + a^2 + a^2 = 3a^2$$

Problem 39: find the shortest distance from the origin to the plane $3x + 2y + z = 12$.

Minimise $f = x^2 + y^2 + z^2$ with $\varphi = 3x + 2y + z - 12 = 0$. $2x + 3\lambda = 0$, $2y + 2\lambda = 0$, $2z + \lambda = 0 \Rightarrow x = -3\lambda/2$, $y = -\lambda$, $z = -\lambda/2$. Substituting: $3(-3\lambda/2) + 2(-\lambda) - \lambda/2 = 12 \Rightarrow -7\lambda = 12 \Rightarrow \lambda = -12/7$.

$x = 18/7$, $y = 12/7$, $z = 6/7$; $f = (324 + 144 + 36)/49 = 504/49 = 72/7$.

$$\text{shortest distance} = \sqrt{(72/7)} = 12/\sqrt{14} \approx 3.207$$

Problem 40: find the maximum volume of a rectangular box inscribed in the sphere $x^2 + y^2 + z^2 = a^2$.

Vertex (x, y, z) in the first octant; $V = 8xyz$. $F = xyz + \lambda(x^2 + y^2 + z^2 - a^2)$: $yz + 2\lambda x = 0$, $zx + 2\lambda y = 0$, $xy + 2\lambda z = 0$. Multiply by x, y, z: $xyz = -2\lambda x^2 = -2\lambda y^2 = -2\lambda z^2 \Rightarrow x = y = z = a/\sqrt{3}$.

$$V_{max} = 8\,(a/\sqrt{3})^3 = \frac{8a^3}{3\sqrt{3}}$$

Problem 41: find the dimensions of an open rectangular box of volume 32 cm^3 with the least surface area.

$S = xy + 2z(x + y)$ with $xyz = 32$. $F = xy + 2yz + 2zx + \lambda(xyz - 32)$. $y + 2z + \lambda yz = 0$, $x + 2z + \lambda xz = 0$, $2y + 2x + \lambda xy = 0$. Multiply the 1st by x, 2nd by y, 3rd by z: $xy + 2xz + \lambda xyz = yx + 2yz + \lambda xyz \Rightarrow x = y$; and with the 3rd: $2xz + \ldots \Rightarrow x = y = 2z$.

$(2z)(2z)z = 32 \Rightarrow z^3 = 8 \Rightarrow z = 2$, $x = y = 4$. Least surface area $S = 16 + 2 \cdot 2 \cdot 8 = 48\ cm^2$.

Problem 42: find the largest volume of a rectangular parallelepiped inscribed in the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$.

$V = 8xyz$; $F = xyz + \lambda(x^2/a^2 + y^2/b^2 + z^2/c^2) \Rightarrow yz + 2\lambda x/a^2 = 0$ etc.

Multiplying by x, y, z and equating: $x^2/a^2 = y^2/b^2 = z^2/c^2 = 1/3 \Rightarrow x = a/\sqrt{3}$, $y = b/\sqrt{3}$, $z = c/\sqrt{3}$.

$$V_{max} = 8 \cdot \frac{abc}{3\sqrt{3}} = \frac{8abc}{3\sqrt{3}}$$

Problem 43: find the points on $z^2 = xy + 1$ nearest to the origin.

Minimise $x^2 + y^2 + z^2$ subject to $z^2 - xy - 1 = 0$. F: $2x - \lambda y = 0$, $2y - \lambda x = 0$, $2z + 2\lambda z = 0$. If $z \neq 0$ then $\lambda = -1 \Rightarrow 2x + y = 0$ and $2y + x = 0 \Rightarrow x = y = 0 \Rightarrow z^2 = 1$. Points $(0, 0, \pm 1)$, minimum distance 1.

Problem 44: maximise $x^2y^3z^4$ subject to $2x + 3y + 4z = a$.

Maximise $\log f = 2\log x + 3\log y + 4\log z$: $2/x + 2\lambda = 0$, $3/y + 3\lambda = 0$, $4/z + 4\lambda = 0 \Rightarrow x = y = z = -1/\lambda$. Constraint: $9x = a \Rightarrow x = y = z = a/9$.

$$f_{max} = (a/9)^2 (a/9)^3 (a/9)^4 = \frac{a^9}{9^9}$$

Important Questions – Unit IV

1. Test continuity of $f(x, y)$ at the origin; explain differentiability.
2. Find partial derivatives; verify $f_{xy} = f_{yx}$; Euler's theorem problems.
3. Total derivative and chain rule problems; implicit differentiation.
4. Directional derivative and direction of maximum rate of change.
5. Taylor / Maclaurin series of a function of two variables.
6. Jacobians; show functional dependence and find the relation.
7. Maxima and minima of functions of two variables ($rt - s^2$ test).
8. Lagrange multipliers: extreme values with one / two constraints; box and distance problems.